PR: 2

Derivable Judgement

A statistical desision making model

PART-A: Theorical Foundation

1. What is inferential stastics ?

Ans: This offers methods to study experiments done on small samples of chalk out the inferences to the entire population.

2. What is hypothesis testing and its components ?

Ans: Hypothesis testing is a statistical method used to make decisions about a population based on sample data.

Components of Hypothesis testing

(i) Null Hypothesis: (H_0)

The null hypothesis is a statement that assume there is no significant difference, effect or relationship between variables

(ii) Alternative Hypothesis (H_1)/(H_a):

The alternative hypothesis is a statement that suggest there is a significant difference, effect, or relationship between variables.

~~3. Find the critical value and p-value to interpret the test results.~~

3. Explain confidence interval and critical value.

Ans: Confideme Interval (CI): It is a range of values that is likely to contain the true population parameter. A 95% CI means that if we repleced the samping many times about 95% of such intervals would contain the true value.

Critical Value: It is the cut-off point on the test dristribution. If the test statistic goes beyond it, H_0 is rejected.

4. Define P-value

Ans: The P-value is the probability of getting a result at least as extreme as the one observed, assuming H_0 is true. A small P-value means the data is unlikely under H_0. If $P < \alpha$ (0.05), we reject H_0; otherwise we fail to reject it.

5. Differentiate Type I and Type II errors.

Ans:

Type I	Type II
→ Rejecting H_0 when it is actually true.	→ Failing to reject H_0 when it is actually false
→ α (equal to the significance level, e.g. 0.05)	→ β (Power of the test $= 1-\beta$)
→ Ex: Concluding smoking affects diabetes when it really does not!	→ Ex: Concluding smoking has no effect when it really does.

6. Brief descriptions of z-test, t-test, chi-square and ANOVA test.

Ans: z-test: Compares a sample mean with a known value when the population standard deviation is known and the sample is large ($n \geq 30$)

t-test: compares means when the population standard deviation is unknown or the sample is small. It uses the t-distribution.

Chi-square test: tests whether two categorical variables are related, by comparing observed counts with expected counts.

ANOVA: tests whether the means of three or more groups differs, using the F-statistic.

7. What is covariance?

Ans: Covariance measures how two variables change together. A positive value means they tend to increase together, a negative value means one increase as the other decreases, and a value near zero means no linear relationship. Its size depends on the units of the variables, so it cannot show the strength of the relationship.

→ Formula: $\text{cov}(X,Y) = \Sigma(x-\bar{x})(y-\bar{y})/(n-1)$

8. What is correlation?

Ans: Correlation measures both the direction and the strengt of a linear relationship between two variables. Person's ranges from −1 (perfect negative) to +1 (perfect positive) and it has no units because it is covariance divide by the product of the standrad deviation.